

IS 340 Exam II

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I. Instructions for Multiple Choice Questions: Please circle the most appropriate choice. You will receive no points if you circle more than one choice (5*8=40 points available).

1. Which of the following statements will produce the value 160?
 - a.
 - b. `SELECT ROUND(155.52, -1) FROM dual;`
 - c.
 - d.
2. Which of the following can be used with date columns?
 - a. c.
 - b. d. all of the above
3. Based upon the contents of the ORDERS table, which of the following will display how many orders were shipped to each state?
 - a. `SELECT shipstate, COUNT(*) FROM orders GROUP BY shipstate;`
 - b.
 - c.
 - d.
4. An outer query is also referred to as a(n) _____.
 - a. c.
 - b. d. parent query
5. The following SQL statement contains what type of subqueries?
`SELECT isbn, title FROM books WHERE pubid =(SELECT pubid FROM books WHERE title = 'SHORTEST POEMS') AND retail-cost >(SELECT AVG(retail-cost) FROM books);`
 - a.
 - b. multiple-row
 - c.
 - d.
6. Which operator will instruct Oracle SQL to list all records with a value that is more than the highest value returned by the subquery?
 - a. c. >ALL
 - b. d.
7. Which of the following is a valid SQL statement?
 - a.
 - b.
 - c. `CREATE TABLE newname (cola DATE, colB NUMBER);`
 - d.
8. Which of the following commands will increase the size of the CITY column in the CUSTOMERS table from 12 to 20 and increase size of the LASTNAME column from 10 to 14?
 - a.
 - b. `ALTER TABLE customers MODIFY (city VARCHAR2(20), lastname VARCHAR2(14));`
 - c.
 - d.

II. Instructions for SQL (or PL/SQL) Statement Questions: Please PRINT your SQL statements. You will not receive any points if I cannot read your writing, and you MAY get partial credits if your answer is incorrect but readable. You may want to run your SQL statements in Oracle SQL Plus to check the query results (12*5=60 points available).

1. Display the order# and the shipment status (name this column as “Status”) of each order. The shipment status of those orders that have not been shipped should be “Not Shipped”, and the status of those shipped orders should be “Shipped”.

(Please print your SQL statement here):

```
SELECT order#,  
CASE  
WHEN shipdate IS NULL THEN 'Not Shipped'  
ELSE 'Shipped'  
END AS Status  
FROM orders;
```

2. How many months are there between today and 12/14/2001? Please round the number of months to the nearest integer (whole number).

(Please print your SQL statement here):

```
SELECT ROUND(MONTHS_BETWEEN(SYSDATE, TO_DATE('14-DEC-2001','DD-MON-YYYY'))) AS  
months_diff  
FROM dual;
```

How many months does your query return? 287 months

3. Use one SQL statement with subqueries to display the category name and the average book retail price of the book category that has the lowest average book retail price.

(Please print your SQL statement here):

```
SELECT category, AVG(retail) AS avg_price  
FROM books  
GROUP BY category  
HAVING AVG(retail) = (  
SELECT MIN(AVG(retail))  
FROM books  
GROUP BY category);
```

What's the category name? Cooking

4. Use one SQL statement with subqueries to display the title and the publisher's name of the book that generates the highest total profits from all orders stored in the database.

```
SELECT b.title, p.name AS publisher  
FROM books b  
JOIN publishers p ON b.pubid = p.pubid  
WHERE b.isbn = (  
SELECT isbn  
FROM (  
SELECT o.isbn, SUM((b.retail - b.cost) * o.quantity) AS total_profit  
FROM orderitems o  
))
```

The book's title is _____, and its publisher's name is _____.

5. Write an anonymous PL/SQL block *using a **WHILE LOOP** statement* to display the value of Y to your screen, where $Y = 1*1 + 2*2 + 3*3 + 4*4 + \dots + 98*98 + 99*99 + 100*100$.

(Please print your PL/SQL block here):

```
DECLARE
  Y NUMBER := 0;
  i NUMBER := 1;
BEGIN
  WHILE i <= 100 LOOP
    Y := Y + i*i;
    i := i + 1;
  END LOOP;
  DBMS_OUTPUT.PUT_LINE('Y = ' || Y);
END;
```

What is the value of Y ? 338350